

3 hours on examples

Problems 1 (97IMO5)

We claim the answers are $(1, 1), (16, 2), (27, 3)$ work.

$$a^{b^2} = b^a \Leftrightarrow b = a^{\frac{b^2}{a}}$$

Let $\frac{b^2}{a} = k$. We have,

$$\Leftrightarrow b^2 = a^{2k} \Leftrightarrow k = a^{2k-1}$$

If $k = \frac{1}{2}$, then $\frac{1}{2} = a^0$ is impossible. Therefore, $k \neq \frac{1}{2}$ and we may,

$$\Leftrightarrow k^{\frac{1}{2k-1}} = a$$

Case 1: $\frac{1}{2} < k < 1$.

On this range, note that $\frac{1}{2k-1} > 0$ but $k < 1$. Therefore, $k^{\frac{1}{2k-1}} < 1 \Leftrightarrow k < 1^{2k-1}$ is true. As a result, $a < 1$ which is impossible as a is a positive integer.

Case 2: $k = 1$

We have $a = 1^{\frac{1}{2-1}} = 1$. Since $\frac{b^2}{a} = k$, we get $b = 1$ as well.

Case 3: $k > 1$

For a given prime p , note that $v_p(k)$ is an integer since $\frac{b^2}{a}$ is a rational number. Additionally, since $v_p\left(k^{\frac{1}{2k-1}}\right) = v_p(a)$ and a is a non-negative integer, we have,

$$\frac{1}{2k-1} \left(v_p(k) \right) = v_p(a)$$

However, $2k-1 \geq k \geq p^{v_p(k)} > v_p(k)$. Therefore, $1 > \frac{1}{2k-1} \left(v_p(k) \right) = v_p(a)$

Clearly, $v_p(a) = 0$ for all primes p meaning $a = 1$. We once again get $b = 1$ also.

Case 4: $k < \frac{1}{2}$

We let $x = \frac{1}{k} = \frac{a}{b^2}$. We may similarly check $b^2 \mid a$ so x is an integer and $x > 2$. We have,

$$a^{b^2} = b^a \Leftrightarrow a = b^x \Leftrightarrow x = b^{x-2}$$

Showing $x \leq 4$ then caseworking suffices.

Problem 3 (22AMO4)

The only solution is $p = 3; q = 2$.

We let $m^2 = p - q; n^2 = pq - q$. Note that since $n^2 = q(p - 1)$ and q is prime, we may write $p = qx^2 + 1$. We have $n = qx$ in this case. Note that,

$$n^2 - m^2 = (n - m)(n + m) = p(q - 1)$$

Since p is prime, we have $p \mid n - m$ or $p \mid n + m$. However,

$$0 < n - m < n = qx < qx^2 + 1 = p$$

Therefore, it is impossible for $p \mid n - m$.

$$p \mid n + m$$

$$\Rightarrow n + m \geq p$$

$$\Leftrightarrow qx + \sqrt{p - q} \geq p$$

$$\Leftrightarrow qx + \sqrt{qx^2 - q + 1} \geq qx^2 + 1$$

$$\Leftrightarrow \sqrt{qx - q + 1} \geq qx^2 - qx + 1$$

Let $qx - q + 1 = y$.

$$\Leftrightarrow \sqrt{y} \geq xy - x + 1$$

$$\Rightarrow y \geq xy - x + 1$$

$$\Leftrightarrow y(1 - x) \geq 1 - x$$

We either have $x = 1$ or $y = 1$. If $y = 1$, we have $qx - q = 0$ so $q = 0$ or $x = 1$. Since $q \neq 0$, we only need to consider $x = 1$ and $p = q + 1$. The only consecutive primes are $(q, p) = (2, 3)$ as desired.

10 clubs; 5.5 hours

Problem 5 (10SLN1)

We claim the answer is $n = 39$. This is achieved with,

$$s_i = i + 1 \text{ for } 1 \leq i \leq 32; s_i = i + 2 \text{ for } 33 \leq i \leq 38; s_{39} = 67$$

Therefore, the product is,

$$\left(\frac{1}{2} \cdot \frac{2}{3} \cdot \dots \cdot \frac{32}{33}\right) \cdot \left(\frac{34}{35} \cdot \frac{35}{36} \cdot \dots \cdot \frac{39}{40}\right) \cdot \left(\frac{66}{67}\right) = \frac{1}{33} \cdot \frac{17}{20} \cdot \frac{66}{67} = \frac{17}{670} = \frac{51}{2010}$$

To show that this is optimal, note that $s_1 \geq 2$ since $s_i = 1$ would result in the product being positive. Therefore, $s_2 \geq 3$; $s_3 \geq 4$; ... Overall,

$$\left(1 - \frac{1}{s_1}\right) \dots \left(1 - \frac{1}{s_n}\right) \geq \left(1 - \frac{1}{2}\right) \left(1 - \frac{1}{3}\right) \dots \left(1 - \frac{1}{n+1}\right) = \frac{1}{n+1}$$

$n \leq 38$ is impossible as $\frac{51}{2010} < \frac{1}{39}$.

Problem 8 (20CYBER2)

We induct on n .

Base case: We have $f(1) = 4$ which has exactly one prime divisor.

Inductive step: FTSOC, let $f(k)$ gain two prime divisors that do not divide $f(1), \dots, f(k-1)$. In other words, for two primes p, q , we have $pq \mid f(k)$ but $p, q \nmid f(1), \dots, f(k-1)$.

Note that, if $\frac{p}{2} < k < p$, then $p \mid 3k^2 + 1 \Leftrightarrow p \mid 3(p-k)^2 + 1$ so $p \mid f(p-k)$ an earlier term. This is impossible.

If $p = k$, then $p \mid 3k^2 + 1$ is clearly impossible.

If $k > p$, then $p \mid 3k^2 + 1 \Leftrightarrow p \mid 3(k-p)^2 + 1$ so $p \mid f(p-k)$ an earlier term. This is also given to be false.

Therefore, $k \leq \frac{p}{2}$. Analogously, $k \leq \frac{q}{2}$. We have,

$$\begin{aligned} pq &\mid 3k^2 + 1 \\ \Leftrightarrow pq &\leq 3k^2 + 1 \\ \Rightarrow pq &\leq \frac{3pq}{4} + 1 \\ \Leftrightarrow pq &\leq 4 \end{aligned}$$

This is clearly impossible.

Since $f(k)$ adds at most one new prime divisor, the inductive hypothesis finishes.

16 clubs; 7.5 hours

Problem 9 (07ROU)

Using the quadratic formula, we must have,

$$x = \frac{(a^2 + b^2 + c^2 + d^2 + 1) \pm \sqrt{(a^2 + b^2 + c^2 + d^2 + 1)^2 - 4(a + c)(b + d)}}{2}$$

For x to be a positive integer, $(a^2 + b^2 + c^2 + d^2 + 1)^2 - 4(a + c)(b + d)$ must be a square.

We write, for some $-1 \leq k \leq a^2 + b^2 + c^2 + d^2$,

$$(a^2 + b^2 + c^2 + d^2 + 1)^2 - 4(a + c)(b + d) = (a^2 + b^2 + c^2 + d^2 - k)^2$$

$$\Leftrightarrow (2a^2 + 2b^2 + 2c^2 + 2d^2 + 1 - k)(k + 1) = 4(a + c)(b + d)$$

$$\Leftrightarrow (2k + 2)(a^2 + b^2 + c^2 + d^2) - (k^2 - 1) = 4(a + c)(b + d)$$

By AM-GM, we know $(a + c)(b + d) = ab + ad + bc + cd \leq a^2 + b^2 + c^2 + d^2$,

$$\Rightarrow (2k - 2)(a^2 + b^2 + c^2 + d^2) - (k^2 - 1) \leq 0$$

If $k = 0$, note that $\sqrt{(a^2 + b^2 + c^2 + d^2 + 1)^2 - 4(a + c)(b + d)} = a^2 + b^2 + c^2 + d^2$ will not give integer solutions to x .

If $k = 1$, we have,

$$(2k + 2)(a^2 + b^2 + c^2 + d^2) - (k^2 - 1) = 4(a + c)(b + d)$$

$$\Leftrightarrow 4(a^2 + b^2 + c^2 + d^2) = 4(a + c)(b + d)$$

By AM-GM, we must have $a = b = c = d$. Therefore,

$$\begin{aligned} x &= \frac{(a^2 + b^2 + c^2 + d^2 + 1) \pm \sqrt{(a^2 + b^2 + c^2 + d^2 + 1)^2 - 4(a + c)(b + d)}}{2} \\ &= 1 \text{ or } a^2 + b^2 + c^2 + d^2 \end{aligned}$$

These are 1^2 and $(2a)^2$ respectively as desired.

If $k > 1$,

$$\Leftrightarrow 2(a^2 + b^2 + c^2 + d^2) \leq k + 1$$

However, $k \leq a^2 + b^2 + c^2 + d^2$ so this case is impossible.

Problem 12 (09SLN3)

FTSOC, may only a finite set of primes $p_1, \dots, p_k$ divide some element $f(c)$.

Note that there exists an integer m_i which is the minimum of $v_{p_i}(f(x))$ over all x because $v_{p_i}(f(x))$ is a non-negative integer and the minimum is at most $v_{p_i}(f(1))$.

Claim: If $v_{p_i}(f(x)) = m_i$, then $v_{p_i}(f(x \pm p_i^{m_i+1})) = m_i$ also if f is defined there.

Proof: FTSOC, if $v_{p_i}(f(x \pm p_i^{m_i+1})) > m_i$ instead, then,

$$\begin{aligned}(x \pm p_i^{m_i+1}) - x &\mid f(x \pm p_i^{m_i+1}) - f(x) \\ \Leftrightarrow p_i^{m_i+1} &\mid f(x \pm p_i^{m_i+1}) - f(x) \\ \Leftrightarrow p_i^{m_i+1} &\mid f(x) \quad (!)\end{aligned}$$

Define $x_1, \dots, x_k$ where each x_i is an example $v_{p_i}(f(x_i)) = m_i$ achieving the minimum.

Using the claim, for all $y_i \equiv x_i \pmod{p_i^{m_i+1}}$, we have $v_{p_i}(f(y_i)) = m_i$. By the CRT, there exists some integer x such that for all y satisfying $y \equiv x \pmod{p_1^{m_1+1} \dots p_k^{m_k+1}}$, we have $y \equiv x_i \pmod{p_i^{m_i+1}}$ for all i . Therefore, we have $v_{p_i}(f(y)) = m_i$ for all y . Since no other primes divide $f(y)$, such $f(y)$ satisfy $f(y) = p_1^{m_1} \dots p_k^{m_k}$.

Since f is non-constant, there exists some $f(z) \neq p_1^{m_1} \dots p_k^{m_k}$.

$$\begin{aligned}z - y &\mid f(z) - f(y) \\ \Rightarrow |f(z) - f(y)| &\geq |z - y|\end{aligned}$$

Taking $y \equiv x \pmod{p_1^{m_1+1} \dots p_k^{m_k+1}}$ large, $f(y)$ remains to be $p_1^{m_1} \dots p_k^{m_k}$ meaning while $|z - y|$ is unbounded meaning no finite value of $f(z)$ may exist. (!)

24 clubs; 10 hours

Problem 15 (Z190706A)

We claim $|a + b\sqrt{2}| \equiv \text{abs}(a + b\sqrt{2})^{\frac{1}{2017}}$ or $|a + b\sqrt{2}| \equiv \text{abs}(a - b\sqrt{2})^{\frac{1}{2017}}$. Both clearly work since $a - b\sqrt{2} = 0$ is impossible for a, b quotients and we may check the inequality.

For any positive integer n , we take $|-n||-n| = |n^2| = n^{\frac{2}{2017}}$. Since $|n| > 0$, we must have $|-n| = n^{\frac{1}{2017}} = |n|$.

For any rational $q = \frac{a}{b}$ for integers a, b , we take $|q||b| = |a|$ to get $|q| = \text{abs}(q)^{\frac{1}{2017}}$. Similarly, for any rational multiplied by square root of two like $q\sqrt{2}$, we may take $|q\sqrt{2}||q\sqrt{2}| = |2q^2| = (2q^2)^{\frac{1}{2017}}$ to get $|q\sqrt{2}| = (\text{abs}(q\sqrt{2}))^{\frac{1}{2017}}$ also.

We may now look at the value of $|3 + 2\sqrt{2}|$.

Case 1: $|3 + 2\sqrt{2}| = 1$.

Note that,

$$1 = |3 + 2\sqrt{2}||3 - 2\sqrt{2}|$$

Therefore, $|3 + 2\sqrt{2}| = |3 - 2\sqrt{2}| = 1$. For any n , $|(3 + 2\sqrt{2})^n| = |(3 - 2\sqrt{2})^n| = 1$ also. However, note that $(3 + 2\sqrt{2})^n$ and $(3 - 2\sqrt{2})^n$ sum to a constant $A_n > 3^n$ by binomial expansion.

$$\begin{aligned} |(3 + 2\sqrt{2})^n + (3 - 2\sqrt{2})^n| &\leq |(3 + 2\sqrt{2})^n| + |(3 - 2\sqrt{2})^n| \\ &\Leftrightarrow A_n^{\frac{1}{2017}} \leq 2 \end{aligned}$$

This is clearly impossible taking n large.

Case 2: $|3 + 2\sqrt{2}| > 1$.

We have $|3 - 2\sqrt{2}| = c < 1$ in this case. We write, for each N , quotients A_N, B_N such that,

$$(3 - 2\sqrt{2})^N = A_N - B_N\sqrt{2}$$

Since the LHS approaches zero as N gets large, $\frac{A_N}{B_N}$ approaches $\sqrt{2}$. We now write, for any quotients x, y ,

$$\begin{aligned} |B_N x + B_N y \sqrt{2}| &\leq |B_N x + A_N y| + |B_N y \sqrt{2} - A_N y| \\ \Leftrightarrow B_N^{\frac{1}{2017}} |x + y \sqrt{2}| &\leq |B_N x + A_N y| + y^{\frac{1}{2017}} |A_N - B_N \sqrt{2}| \\ \Leftrightarrow B_N^{\frac{1}{2017}} |x + y \sqrt{2}| &\leq |B_N x + A_N y| + y^{\frac{1}{2017}} c^N \\ \Leftrightarrow B_N^{\frac{1}{2017}} |x + y \sqrt{2}| &\leq \text{abs}(B_N x + A_N y) + y^{\frac{1}{2017}} c^N \end{aligned}$$

Note that $B_N x + A_N y$ is a quotient,

$$\Leftrightarrow |x + y\sqrt{2}| \leq \text{abs}\left(x + \frac{A_N}{B_N}y\right) + \frac{y^{\frac{1}{2017}}c^N}{B_N^{\frac{1}{2017}}}$$

Taking N large, since $c < 1$ and $\frac{A_N}{B_N}$ approaches $\sqrt{2}$,

$$\Leftrightarrow |x + y\sqrt{2}| \leq \text{abs}\left(x + \frac{A_N}{B_N}y\right)$$

Analogously,

$$|B_Nx + B_Ny\sqrt{2}| \geq |B_Nx + A_Ny| - |A_Ny - B_Ny\sqrt{2}|$$

$$\Leftrightarrow |x + y\sqrt{2}| \geq \text{abs}\left(x + \frac{A_N}{B_N}y\right) - \frac{y^{\frac{1}{2017}}c^N}{B_N^{\frac{1}{2017}}}$$

$$\Leftrightarrow |x + y\sqrt{2}| \geq \text{abs}\left(x + \frac{A_N}{B_N}y\right)$$

In this case, as desired, $|x + y\sqrt{2}| = \text{abs}\left(x + \frac{A_N}{B_N}y\right)^{\frac{1}{2017}}$

Case 3: $|3 + 2\sqrt{2}| = c < 1$

If we let $(3 + 2\sqrt{2})^N = A_N + B_N\sqrt{2}$, we once again have $\frac{A_N}{B_N}$ approaches $\sqrt{2}$.

$$|B_Nx + B_Ny\sqrt{2}| \leq |B_Nx - A_Ny| + |A_Ny + B_Ny\sqrt{2}|;$$

$$|B_Nx + B_Ny\sqrt{2}| \geq |B_Nx - A_Ny| - |A_Ny + B_Ny\sqrt{2}|$$

We may get the similar conclusion that $|x + y\sqrt{2}| = (x - y\sqrt{2})^{\frac{1}{2017}}$

Problem 17 (08SLN2)

WLOG, let $a_1 < a_2 < \dots < a_n$. FTSOC, let every pair of distinct a_i, a_j satisfy $a_i + a_j \mid a_k$ for some k . By the well-ordering principle, there must be a solution that minimizes the quantity $a_1 + \dots + a_n$.

Note that each $3a_i$ has at most two divisors greater than a_i which are $3a_i$ and possibly $\frac{3a_i}{2}$.

Out of $a_n + a_{n-1}, a_n + a_{n-2}, \dots, a_n + a_1$, we consider what they may divide. Since $a_n + a_i > a_k$ for all k , we have,

$$a_n + a_i \mid a_k \Rightarrow a_n + a_i \in \left\{ \frac{3}{2}a_k, 3a_k \right\}$$

Therefore, $3 \mid a_n + a_i$ always for $1 \leq i \leq n-1$.

If $a_n \equiv 0 \pmod{3}$, then $a_i \equiv 0 \pmod{3}$ for all a_i . We may divide each term by three to get another sequence with smaller sum still satisfying $a_i + a_j \mid a_k$ for some k for all distinct a_i, a_j . (!)

If $a_n \equiv 1, 2 \pmod{3}$, then $a_i \equiv 2, 1 \pmod{3}$ respectively. No other pairs $a_i, a_j < a_n$ sum to a multiple of three. Among these pairs, there hence exists an a_k such that,

$$a_i + a_j \mid 3a_k \Leftrightarrow a_i + a_j \mid a_k$$

Now consider $a_{n-1} + a_i \mid a_k$ for $1 \leq i \leq n-2$. Since $a_{n-1} + a_i > a_{n-1}$, it is impossible for $k \leq n-1$. Therefore, $a_{n-1} + a_i \mid a_n$ for $1 \leq i \leq n-2$.

The greatest proper divisor of a_n is at most $\frac{a_n}{2}$. However, if $a_{n-1} + a_i \leq \frac{a_n}{2}$, then $a_{n-1} < \frac{a_n}{2}$.

However, we previously had $a_n + a_{n-1} \in \left\{ 3a_k, \frac{3}{2}a_k \right\}$ for some k .

k may not be n since $a_{n-1} < \frac{a_n}{2}$ means $a_n + a_{n-1} < \frac{3}{2}a_n$ in that case. k may not be $n-1$ or less since $a_n + a_{n-1} > 3a_{n-1}$. Therefore, for all i , the only option is,

$$a_{n-1} + a_i = a_n$$

This is impossible for $n > 3$ since a_i is distinct.

For $n = 3$, we let $a_1 = x; a_2 = y$ meaning $a_3 = x + y$. We know $a_2 + a_3 = x + 2y$ may not divide $3x$ by bounding as $y > x$. It also may not divide $3y$ since $x + 2y \mid 3y \Leftrightarrow x + 2y \mid y - x$ is also untrue by bounding. Therefore, $x + 2y \mid 3x + 3y \Leftrightarrow x + 2y \in \left\{ 3x + 3y, \frac{3x+3y}{2} \right\}$. It is impossible for either of these to happen by bounding.

32 clubs; 15 hours

Problem 18 (15AMO5)

WLOG, swap $(a, b) \leftrightarrow (c, d)$ as necessary such that $a < c$. Note that we may do this since this does not affect $ac + bd$. Since $a^4 + b^4 = c^4 + d^4$, we know $a < c \Rightarrow b > d$.

FTSOC, let $p = ac + bd$. Note that,

$$p \mid ac + bd$$

$$\begin{aligned}
&\Leftrightarrow a^4 c^4 \equiv b^4 d^4 \pmod{p} \\
&\Leftrightarrow a^4 (e^5 - d^4) \equiv (e^5 - a^4) d^4 \pmod{p} \\
&\Leftrightarrow p \mid e^5 (a - d)(a + d)(a^2 + d^2)
\end{aligned}$$

Clearly, $ac + bd > a - d$ and $a + d$ since b, c are distinct positive integers. We also have $ac + bd > a \cdot a + d \cdot d$ since $c > a$ and $b > d$. Therefore, p may not divide $a - d, a + d$, or $a^2 + d^2$.

$$\begin{aligned}
&\Leftrightarrow p \mid e^5 \Leftrightarrow p \mid e \\
&\Rightarrow ac + bd \leq e \\
&\Leftrightarrow (ac + bd)^5 \leq e^5 \\
&\Leftrightarrow (ac + bd)^5 \leq a^4 + b^4
\end{aligned}$$

This is clearly untrue since $a^5 c^5 > a^4$ and $b^5 d^5 > b^4$ already.

Problem 20 (14BGR4)

(I was staring at the screen while doing 15AM05 clueless on what to do and head-solved this so might as well :/)

We claim the only solutions are $(2, 3)$ and $(3, 2)$. We can check that both work.

We look for any other solutions (p, q) . Note that since $(2, 3)$ are the only twin primes and $p = q$ is clearly impossible, we must have $|p - q| \geq 2$.

$$\begin{aligned}
p^2 \mid q^3 + 1 &\Leftrightarrow p^2 \mid (q + 1)(q^2 - q + 1) \\
q^2 \mid p^6 - 1 &\Leftrightarrow q^2 \mid (p - 1)(p + 1)(p^2 + p + 1)(p^2 - p + 1)
\end{aligned}$$

Case 1: $p > q$

In this case, since $p - q \geq 2$, it is impossible for $p \mid q + 1$. We must then have,

$$p^2 \mid q^2 - q + 1$$

This is also impossible for $p > q$ by bounding.

Case 2: $p < q$

Once again, it is impossible for $q \mid p - 1$ or $q \mid p + 1$. By bounding, it is impossible for $q^2 \mid p^2 + p + 1$ or $q^2 \mid p^2 - p + 1$ since $q \geq p + 2$ so $q \mid p^2 + p + 1$ and $q \mid p^2 - p + 1$. Taking the difference, $q \mid 2p$. This is impossible. (!)

39 clubs; 17.5 hours

Problem 21 (23IMO3)

Claim 1: The sequence is non-decreasing.

Proof: FTSOC let the sequence have a pair of consecutive decreasing elements. Consider whenever we have $a_{i-2} \leq a_{i-1} > a_i$ for some i . Since P only has non-negative coefficients,

$$a_{i-1} > a_i \Rightarrow f(a_{i-1}) > f(a_i) \Rightarrow a_i > a_{i+k}$$

However, similarly,

$$a_{i-2} \leq a_{i-1} \Rightarrow a_{i-1} \leq a_{i+k-1}$$

Therefore, since,

$$a_{i+k-1} \geq a_{i-1} > a_i > a_{i+k}$$

We have a new pair of decreasing consecutive terms (a_{i+k-1}, a_{i+k}) . We may induct to see that all pairs (a_{i+jk-1}, a_{i+jk}) for some non-negative integer j are also decreasing.

Additionally, $a_i > a_{i+k} > a_{i+2k} > a_{i+3k} > \dots$ which is impossible since $\{a_n\}$ are positive integers. This case is not possible.

Since the sequence still must have a pair of decreasing elements, we must have, for some integer $m \geq 2$,

$$a_1 > a_2 > a_3 > \dots > a_m \leq a_{m+1} \leq a_{m+2} \leq \dots$$

In this case,

$$a_{m-1} > a_m \Rightarrow a_m > a_{m+k}$$

This is also impossible as the sequence is non-decreasing after a_m . (!)

Claim 2: Either the sequence is entirely constant or entirely increasing.

Sketch of proof (ineligible for bounty claims): If there is some $a_i = a_{i+1}$, induct in both directions using $a_i = a_{i+1} \Leftrightarrow a_{i+1} = a_{i+k}$. Use fact that sequence is non-decreasing. $\square$

[The following section is based on official solution since I got stuck]

Claim 3: There exists some C such that $a_{n+k} - a_n \leq C$.

Proof: We claim that taking a sufficiently large constant C such that $x^k + Cx^{k-1} \geq P(x)$ for all x works. Indeed, for all n ,

$$a_{n+k} = \frac{P(a_n)}{a_n a_{n+1} a_{n+2} \dots a_{n+k-1}} \leq \frac{P(a_n)}{a_n^{k-1}} \leq \frac{a_n^k + C a_n^{k-1}}{a_n^{k-1}} = a_n + C \quad \square$$

Consider a sequence of tuples of k integers (i.e. tuple<int, int, ..., int>) $\{\Delta_n\}$ where,

$$\Delta_n = (a_{n+1} - a_n, a_{n+2} - a_{n+1}, \dots, a_{n+k} - a_{n+k-1})$$

Since each element of Δ_n is non-negative and the sum is at most C , by stars and bars, the number of possible elements for Δ_n is finite. As a result, some tuple $(d_1, d_2, \dots, d_k)$ must be repeated infinitely many times in Δ_n . Each time this element appears in Δ_x , we have, since P is a degree- k polynomial,

$$P(a_x) = (a_x + d_1)(a_x + d_1 + d_2) \dots (a_x + d_1 + \dots + d_k)$$

Since this is true for infinitely many a_x as a_x is increasing, in general, $P(x) = (x + d_1)(x + d_1 + d_2) \dots (x + d_1 + \dots + d_k)$. It is hence impossible for two distinct tuples to repeat infinitely many times as $P(x)$ cannot have two definitions and the sequence $d_1, d_1 + d_2, \dots, d_1 + \dots + d_k$ being increasing.

If the elements of the repeated tuple $(d_1, d_2, \dots, d_k)$ are not all the same, then $\Delta_{n+1} = (d_2, \dots, d_k, x)$ for some x with none of them equal to $(d_1, d_2, \dots, d_k)$. Since $(d_2, \dots, d_k, x)$ for some x must appear infinitely many times and x comes from a finite set of values, one of these tuples will repeat infinitely many times which is not allowed. Therefore, in the repeating tuple $(d_1, d_2, \dots, d_k)$ we have $d_1 = d_2 = \dots = d_k := d$.

In general,

$$P(x) = (x + d)(x + 2d) \dots (x + kd)$$

Additionally, $\{\Delta_n\}$ must eventually be constant at $(d, d, d, \dots, d)$. We induct backwards from the threshold that Δ_n becomes constant to finish.

48 clubs; 20 hours